Technological University of the Philippines Visayas
COLLEGE OF ENGINEERING
Talisay City, Negros Occidental

Parallel Communication with CPLDs

A Project presented to the

Computer Engineering Department

of Technological University of the Philippines Visayas

In Partial Fulfillment

of the Requirements for the Subject

Computer Programming 6

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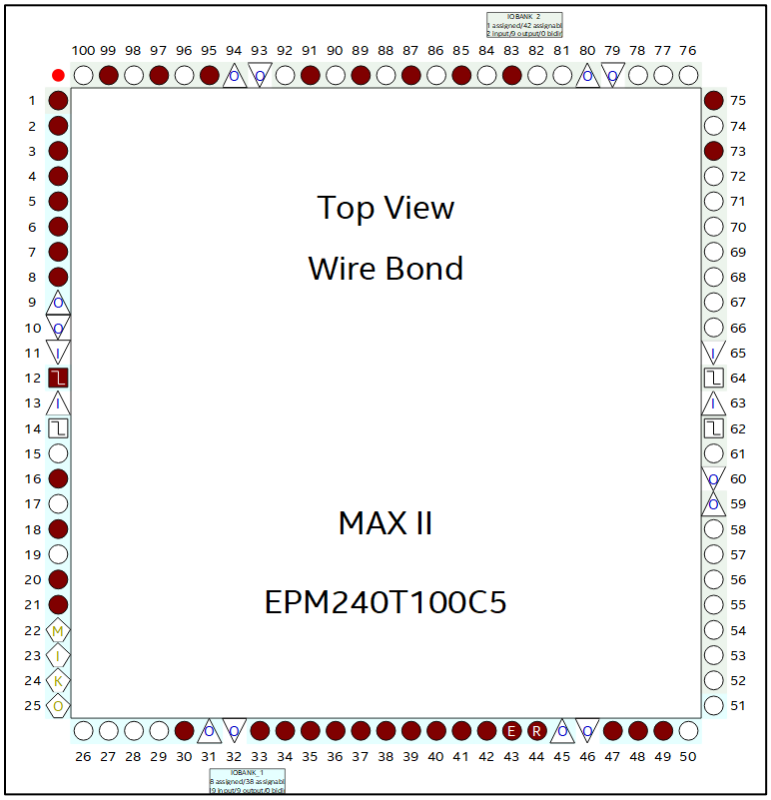
Verilog Code

```
module UART_Test (dataLedOut,dataTx,drdTx,drcRx,dataSwIn,dataRxIn,drcTx,drdRx,load_mode,clk);

   output reg[7:0] dataLedOut,dataTx;
   output reg drdTx,drcRx;
   input[7:0] dataSwIn,dataRxIn;
   input[1:0] load_mode;
   input drdRx,drcTx,clk;

   reg[23:0] counter;
   reg[7:0] dataToCount;

   initial begin
      counter = 0;
      dataToCount = 0;
   end

   always @(posedge clk) begin

      case(load_mode)

         // Counter
         2'b00: begin

            counter <= counter + 1;

            if (counter > 16000000) begin
               dataToCount <= dataToCount + 1;
               dataLedOut <= dataToCount;
               counter <= 0;
            end

         end

         // Counter
         2'b01: begin

            counter <= counter + 1;

            if (counter > 16000000) begin
               dataToCount <= dataToCount + 1;
               dataLedOut <= dataToCount;
               counter <= 0;
            end

         end

         // TX Mode
         2'b10: begin

            dataLedOut <= 0;
            drcRx <= 0;
            drdTx <= 1;

            if(drdTx) begin
               dataTx <= dataSwIn;
            end

            if(drcTx) begin
               drdTx <= 0;
            end

         end

         // RX Mode
         2'b11: begin

            drdTx <= 0;

            if(drdRx == 1) begin
               dataLedOut <= dataRxIn;
               drcRx <= 1;
            end

         end

      endcase

   end

endmodule
```

Pin Planner



Node Name	Direction	Location	I/O Bank	:itter Locatior	I/O Standard	Reserved	ırrent Strengt	ict Preservati
clk	Input	PIN_12	1	PIN_12	3.3-V LVTTL		16mA ...ault)	
dataLedOut[7]	Output	PIN_99	2	PIN_99	3.3-V LVTTL		16mA ...ault)	
dataLedOut[6]	Output	PIN_97	2	PIN_97	3.3-V LVTTL		16mA ...ault)	
dataLedOut[5]	Output	PIN_95	2	PIN_95	3.3-V LVTTL		16mA ...ault)	
dataLedOut[4]	Output	PIN_91	2	PIN_91	3.3-V LVTTL		16mA ...ault)	
dataLedOut[3]	Output	PIN_89	2	PIN_89	3.3-V LVTTL		16mA ...ault)	
dataLedOut[2]	Output	PIN_87	2	PIN_87	3.3-V LVTTL		16mA ...ault)	
dataLedOut[1]	Output	PIN_85	2	PIN_85	3.3-V LVTTL		16mA ...ault)	
dataLedOut[0]	Output	PIN_83	2	PIN_83	3.3-V LVTTL		16mA ...ault)	
dataRxIn[7]	Input	PIN_30	1	PIN_30	3.3-V LVTTL		16mA ...ault)	
dataRxIn[6]	Input	PIN_34	1	PIN_34	3.3-V LVTTL		16mA ...ault)	
dataRxIn[5]	Input	PIN_36	1	PIN_36	3.3-V LVTTL		16mA ...ault)	
dataRxIn[4]	Input	PIN_38	1	PIN_38	3.3-V LVTTL		16mA ...ault)	
dataRxIn[3]	Input	PIN_40	1	PIN_40	3.3-V LVTTL		16mA ...ault)	
dataRxIn[2]	Input	PIN_42	1	PIN_42	3.3-V LVTTL		16mA ...ault)	
dataRxIn[1]	Input	PIN_44	1	PIN_44	3.3-V LVTTL		16mA ...ault)	
dataRxIn[0]	Input	PIN_48	1	PIN_48	3.3-V LVTTL		16mA ...ault)	
dataSwIn[7]	Input	PIN_21	1	PIN_21	3.3-V LVTTL		16mA ...ault)	
dataSwIn[6]	Input	PIN_20	1	PIN_20	3.3-V LVTTL		16mA ...ault)	
dataSwIn[5]	Input	PIN_18	1	PIN_18	3.3-V LVTTL		16mA ...ault)	
dataSwIn[4]	Input	PIN_16	1	PIN_16	3.3-V LVTTL		16mA ...ault)	
dataSwIn[3]	Input	PIN_8	1	PIN_8	3.3-V LVTTL		16mA ...ault)	
dataSwIn[2]	Input	PIN_6	1	PIN_6	3.3-V LVTTL		16mA ...ault)	
dataSwIn[1]	Input	PIN_4	1	PIN_4	3.3-V LVTTL		16mA ...ault)	
dataSwIn[0]	Input	PIN_2	1	PIN_2	3.3-V LVTTL		16mA ...ault)	
dataTx[7]	Output	PIN_33	1	PIN_33	3.3-V LVTTL		16mA ...ault)	
dataTx[6]	Output	PIN_35	1	PIN_35	3.3-V LVTTL		16mA ...ault)	
dataTx[5]	Output	PIN_37	1	PIN_37	3.3-V LVTTL		16mA ...ault)	
dataTx[4]	Output	PIN_39	1	PIN_39	3.3-V LVTTL		16mA ...ault)	
dataTx[3]	Output	PIN_41	1	PIN_41	3.3-V LVTTL		16mA ...ault)	
dataTx[2]	Output	PIN_43	1	PIN_43	3.3-V LVTTL		16mA ...ault)	
dataTx[1]	Output	PIN_47	1	PIN_47	3.3-V LVTTL		16mA ...ault)	
dataTx[0]	Output	PIN_49	1	PIN_49	3.3-V LVTTL		16mA ...ault)	
drcRx	Output	PIN_7	1	PIN_7	3.3-V LVTTL		16mA ...ault)	
drcTx	Input	PIN_3	1	PIN_3	3.3-V LVTTL		16mA ...ault)	
drdRx	Input	PIN_5	1	PIN_5	3.3-V LVTTL		16mA ...ault)	
drdTx	Output	PIN_1	2	PIN_1	3.3-V LVTTL		16mA ...ault)	
load_mode[1]	Input	PIN_75	2	PIN_75	3.3-V LVTTL		16mA ...ault)	
load_mode[0]	Input	PIN_73	2	PIN_73	3.3-V LVTTL		16mA ...ault)	

Flow Summary

Flow Status	Successful - Wed Jan 15 11:03:24 2025
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition
Revision Name	UART_Test
Top-level Entity Name	UART_Test
Family	MAX II
Device	EPM240T100C5
Timing Models	Final
Total logic elements	62 / 240 (26 %)
Total pins	39 / 80 (49 %)
Total virtual pins	0
UFM blocks	0 / 1 (0 %)